

Java Debugging

The Art of Code Investigation

codehive

codehive

What is Debugging?

Debugging is the systematic process of identifying, tracing, and fixing errors (bugs) within a computer program. It is an essential skill that transforms a coder into a developer.

Finding Errors: Pinpointing exactly where code deviates from expected behavior.

Reading Logs: Interpreting stack traces and error messages provided by the JVM.

State Tracing: Monitoring how variable values change during step-by-step execution.

codehive

1. The Print Statement Method

The simplest way to debug is to "print" the state of your variables at different points in your code to see where it breaks.

```
int age = 20;
System.out.println("DEBUG: Current age is " + age);

age = age - 25;
System.out.println("DEBUG: Age after calculation: " + age);

if (age < 0) {
    System.out.println("ERROR: Negative age detected!");
}
```

Note: While effective for small scripts, this method can clutter your code in large projects.

2. Professional IDE Debugging

Using an IDE like IntelliJ, Eclipse, or VS Code provides advanced features that make debugging much faster.

Breakpoints: Click next to the line number to tell the program to "pause" right there.

Step Over: Execute the next line of code without entering methods.

Step Into: Enter a method to see exactly what happens inside its logic.

Variables View: A dedicated window that shows the live values of all variables in memory.

3. Debugging Checklist

Follow these steps whenever you encounter a bug:

1. **Reproduce:** Figure out exactly what input causes the error.
2. **Isolate:** Find the specific method or line of code responsible.
3. **Analyze:** Understand *why* the current logic produces the wrong result.
4. **Fix & Test:** Correct the code and try to break it again with new inputs.

Master the Bug. Master the Code.

codehive